Exercise 4

A walker with a body, on a solid floor

At the end, the walker stands in the room with a body that physics can move, on a floor that physics treats as one solid shape.

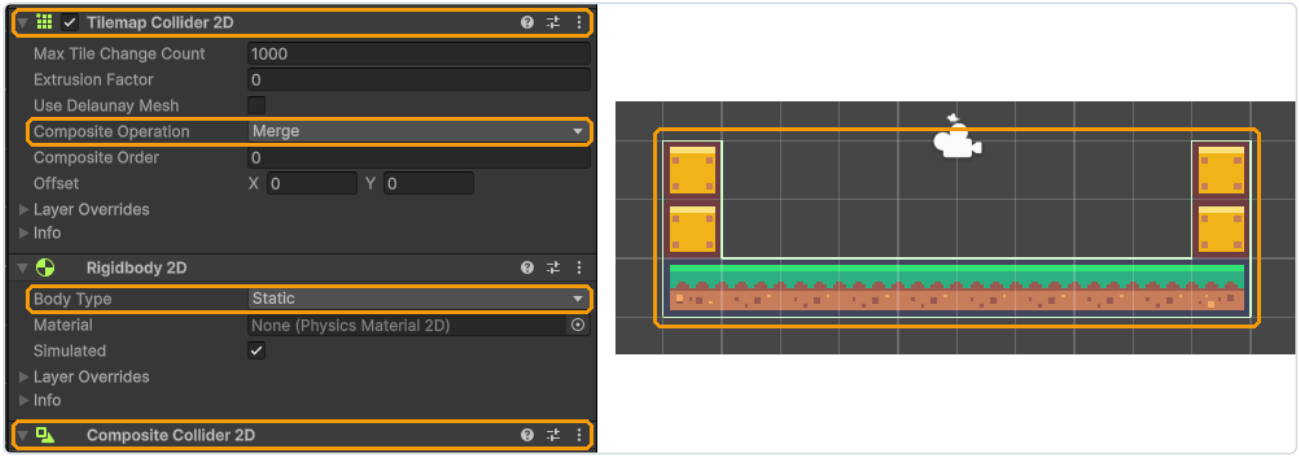
Before you start

- Exercise 3 done and pushed: `MovementLab` with its floor and walls. In a new session, run `git pull` first

1 Floor collision

Three components give `Ground` one solid shape; without Merge every tile keeps its own box, and the walker can snag on the seams.

- On `Ground`: **Add Component**, **Tilemap Collider 2D**, then **Composite Collider 2D**, which brings a Rigidbody 2D.
- Rigidbody 2D: Body Type **Static**, so the floor never moves.
- Tilemap Collider 2D: Composite Operation **Merge**: one green outline now surrounds the floor and the walls.



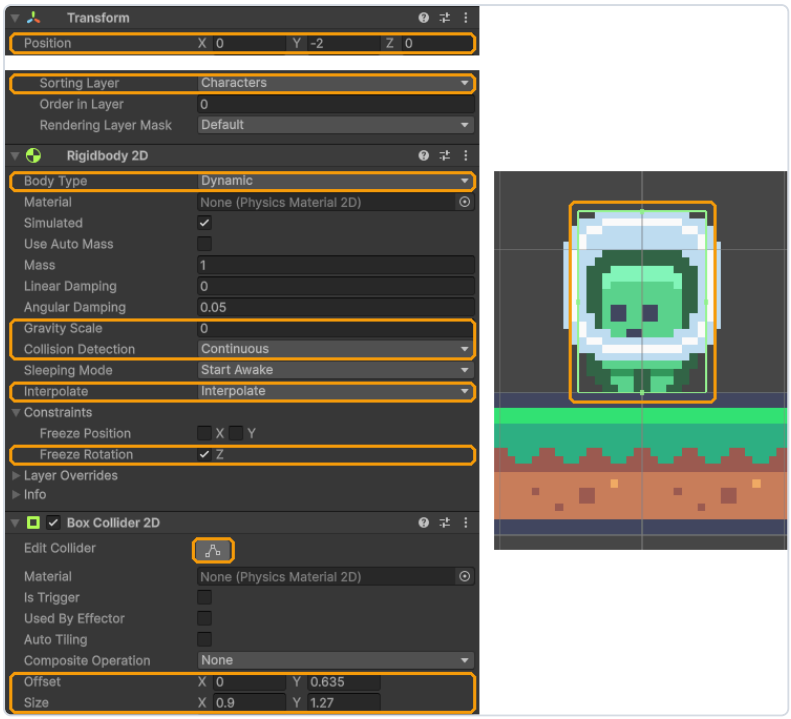
Merge and Static set; one green outline around the whole shape, not a box per tile.

2 The walker

The walker is a sprite with a body: the Rigidbody 2D moves it, and the Box Collider 2D gives it a shape to hit things with.

- Expand `tilemap-characters_packed` in `Assets/Sprites`, drag its frame `_0`, the astronaut standing, into the Hierarchy and rename it `Walker`.
- Set its Position to `(0, -2, 0)`: the floor's top is at y = -2, and with the pivot at its feet, the walker stands on the grass.
- Sprite Renderer: Sorting Layer `Characters`, so it draws in front of the level.
- **Add Component**, **Rigidbody 2D**, and set:
  - Body Type: **Dynamic**
  - Gravity Scale: `0`, as gravity comes in Exercise 6
  - Collision Detection: **Continuous**, so fast moves can't pass through walls
  - Interpolate: **Interpolate**, for smooth motion between physics steps
  - Constraints, Freeze Rotation: **Z**, so it never tips over

- **Add Component**, **Box Collider 2D**: Size `(0.9, 1.27)` and Offset `(0, 0.635)`, a box that hugs the body from the feet up, a little narrower than the helmet. **Edit Collider** shows it on the sprite.



The walker's settings; the green box hugs the body and ends at the feet.

- **Play**: nothing moves and the Console stays empty; stop.
- `Ctrl` + `S`, then commit and push with the message `Add the walker on a solid floor`.